ISPRS Open Journal of Photogrammetry and Remote Sensing 12 (2024) 100062

Contents lists available at ScienceDirect

ISPRS Open Journal of Photogrammetry and Remote Sensing

journal homepage: www.journals.elsevier.com/isprs-open-journal-of-photogrammetry-and-remote-sensing

Deep learning techniques for hyperspectral image analysis in agriculture: A review

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ARTICLE INFO

Keywords:
Hyperspectral imaging
HSI
Deep learning
Agriculture
CNN
RNN
GAN

ABSTRACT

In recent years, there has been a growing emphasis on assessing and ensuring the quality of horticultural and agricultural produce. Traditional methods involving field measurements, investigations, and statistical analyses are labour-intensive, time-consuming, and costly. As a solution, Hyperspectral Imaging (HSI) has emerged as a non-destructive and environmentally friendly technology. HSI has gained significant popularity as a new technology, particularly for its promising applications in remote sensing, notably in agriculture. However, classifying HSI data is highly complex because it involves several challenges, such as the excessive redundancy of spectral bands, scarcity of training samples, and the intricate non-linear relationship between spatial positions and spectral bands. Notably, Deep Learning (DL) techniques have demonstrated remarkable efficacy in various HSI analysis tasks, including those within agriculture. As interest continues to surge in leveraging HSI data for agricultural applications through DL approaches, a pressing need exists for a comprehensive survey that can effectively navigate researchers through the significant strides achieved and the future promising research directions in this domain. This literature review diligently compiles, analyzes, and discusses recent endeavours employing DL methodologies. These methodologies encompass a spectrum of approaches, ranging from Autoencoders (AE) to Convolutional Neural Networks (CNN) (in 1D, 2D, and 3D configurations), Recurrent Neural Networks (RNN), Deep Belief Networks (DBN), Generative Adversarial Networks (GAN), Transfer Learning (TL), Semi-Supervised Learning (SSL), Few-Shot Learning (FSL) and Active Learning (AL). These approaches are tailored to address the unique challenges posed by agricultural HSI analysis. This review evaluates and discusses the performance exhibited by these diverse approaches. To this end, the efficiency of these approaches has been rigorously analyzed and discussed based on the results of the state-of-the-art papers on widely recognized land cover datasets. Github repository.

1. Introduction

Over the past two decades, a growing imperative has been to assess and safeguard the quality of horticultural and agricultural produce. Conventional methods, which include statistical analysis, field measurements, and investigations, have shown to be time-consuming, labour-intensive, and costly (Ravikanth et al., 2017). In response to these limitations, non-destructive and environmentally friendly technologies have gained prominence. Thus, Hyperspectral Imaging (HSI) has emerged as a promising solution. HSI enables capturing multiple images across various wavelengths (Chang, 2003), facilitating precise monitoring of spatial and temporal variations in farmland (Ge et al., 2016). This capability empowers rapid and accurate crop growth predictions and enhances agriculture decision-making.

Hyperspectral imaging (HSI), also termed chemical and spectroscopic imaging, marks a groundbreaking advancement, blending traditional imaging techniques with spectroscopy. This fusion permits the simultaneous collection of spatial and spectral data from an image.

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https://doi.org/10.1016/j.ophoto.2024.100062
Received 28 April 2023; Received in revised form 9 March 2024; Accepted 18 March 2024
Available online 30 March 2024
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While its initial design catered to remote sensing, the advantages of HSI technology in surpassing traditional machine vision capabilities have been recognized in multiple domains, including agriculture and optical sensing. Additionally, the evolution of imaging methodologies has elevated HSI to a pivotal role in the technical evaluation and uniformity measurement of fruits and vegetables.

HSI offers farmers indispensable tools that significantly improve their decision-making capabilities. Its wide-ranging versatility encompasses monitoring crop health (Zhang et al., 2016), growth, and resource allocation, as well as predicting crop yields and detecting diseases (Li et al., 2020a). Additionally, HSI extends its reach to vital tasks such as Land Use Land Cover (LULC) monitoring (Selige et al., 2006), water resource assessment (Gürsoy et al., 2015), and soil condition evaluation (Weber et al., 2008). However, the rich information provided by the HSI technologies comes with a high drawback, where it is almost impossible for humans to analyze a large number of samples. As a result, a growing demand arises for more efficient and automated HSI data analysis in agriculture, marking the trend toward integrating Artificial Intelligence (AI) techniques (Bouguettaya et al., 2022). To address the challenges posed by the extensive data generated by HSI systems, the agricultural sector is increasingly turning to AI, focusing on Deep Learning (DL) approaches. This integration marks a significant leap forward, as AI can efficiently navigate and extract valuable insights from the vast datasets produced by HSI technology. AI, especially DL, offers unparalleled efficiency and precision compared to traditional methods (Pan et al., 2018; Bougourzi et al., 2022), making it an indispensable tool in modern agriculture. As AI and HSI continue to converge, they unlock many possibilities for revolutionizing fresh product quality management (He et al., 2019) and other critical aspects of agriculture, ultimately contributing to a more sustainable and productive agricultural industry.

The primary contribution of this paper is to examine the applications of HSI in the agricultural field, analysing deep learning approaches used for HSI data and evaluating their performance and efficiency for real-world applications. This paper provides insights into new possibilities for developing intelligent systems in horticultural, agricultural, and food domains. It contributes to the advancement of the use of AI in the industry. This paper's main contributions can be summed up as follows:

- Presenting the current state of HSI in agriculture, highlighting significant advancements and pinpointing critical areas needing further research
- Analyzing various publicly available HSI agricultural datasets, highlighting their unique features and the achieved state-of-the-art performance on each dataset.
- Reviewing different techniques and approaches of DL used for HSI, providing an in-depth analysis of their strengths and limitations.
- Presenting and discussing the main applications in the agriculture field that benefit from HSI technology. This section provides insights into how HSI can improve crop yield, quality, and safety.

This paper is arranged as follows: Section 2 discusses a literature landscape. Section 3 summarizes various hyperspectral agricultural datasets. Section 4 analyzes various DL approaches utilized for Agricultural analysis using HSI data. In Section 5, different DL training approaches are discussed. Section 6 discusses the applications of HSI in agriculture and includes topics like anomaly detection, moisture estimation and the plastic pollution issue. The paper concludes with a comprehensive discussion of the limitations and potential methods for future research in sections 7, 8.

2. State-of-the-art

The rapidly evolving landscape of HSI in agriculture has brought significant developments, notably in the early detection of plant diseases and crop health monitoring through advanced imaging technologies and machine learning. It is imperative to identify and address the existing research gaps to hone the capabilities of HSI technologies further and meet the evolving demands of the agriculture sector. Table 1 delineates the findings of various studies that shed light on the current state of HSI applications in agriculture, focusing on the advancements and identifying the pertinent research gaps. This research leaps forward by exploring the applications of HSI in agriculture and bringing DL techniques into the purview. This delves deep into examining the effectiveness of DL methods using prominent land cover datasets and scrutinizes various HSI applications in agriculture. In doing so, we identify the critical research gaps, including the need to identify the most powerful DL techniques and examine the latest uses of these techniques in the agriculture industry.

3. HSI datasets for agriculture analysis

The advent of HSI cameras has ushered in a new era in agricultural research and applications, leading to the collection and labelling of extensive datasets for Agricultural analysis using HSI data. These datasets have significantly expanded the possibilities for machine learning techniques, particularly DL, which requires considerable data for training and evaluation. Table 2 serves as a concise summarization of the available HSI datasets, in which the details of each dataset are highlighted. This table presents crucial details about each dataset, such as its size, spatial resolution, spectral channels, and the number of distinct classes it encompasses.

The Indian Pines dataset (IP) (Scenes, 2020), captured in 1992 using NASA's AVIRIS sensor, offers a balance of size and diversity with 10,249 samples across 16 classes. It boasts a high spectral resolution with 220 bands, making it suitable for detailed spectral analysis. However, its spatial resolution of 20 m is more moderate, positioning it well for broader Land Use Land Cover (LULC) analysis rather than fine-grained studies. This dataset is particularly useful for general agricultural studies and understanding broad LULC patterns. In contrast, the Salinas dataset (Scenes, 2020), also captured by NASA AVIRIS but in 1998, is larger, encompassing 54,129 samples in 16 classes. A very high spectral resolution of 224 bands is exceptional for fine-grained analysis, and the high spatial resolution of 3.7 m makes it ideal for detailed crop classification tasks. Researchers looking for in-depth agricultural class analysis and precise crop studies will find the Salinas dataset particularly beneficial.

The Pavia University dataset (Scenes, 2020), collected in 2001 using the ROSIS-03 sensor, is notable for its large size (42,776 samples) and fewer classes (9). Its moderate spectral resolution of 115 bands is adequate for urban agricultural studies. The dataset's standout feature is its very high spatial resolution of 1.3 m, making it perfect for detailed analysis in urban areas. This dataset is best suited for urban agriculture and detailed land cover studies, especially in smaller, more defined areas. The Botswana (Scenes, 2020) and Chikusei datasets (Yokoya and Iwasaki, 2016), along with the WHU-Hi series (Zhong et al., 2018) (WHU-Hi-HanChuan, WHU-Hi-HongHu, WHU-Hi-LongKou), vary in size and diversity but they are helpful for crop classification tasks. These datasets offer high spectral resolutions (over 240 bands in some cases), facilitating detailed spectral analysis. The spatial resolution varies across these datasets, with some in the WHU-Hi series offering exceptionally high resolutions (ranging from 0.043 to 0.463 m). The WHU-Hi series, in particular, is excellent for high-resolution crop classification and hyperspectral image classification, while the Botswana and Chikusei datasets are more suited for land cover classification and segmentation between urban and rural areas.

When selecting an HSI dataset for an agricultural analysis task, it is important to consider various factors like dataset size and diversity, spectral resolution, spatial resolution, and specific applications. These considerations help align each dataset's strengths with the requirements of the intended research or analysis task. Indeed, the diverse range of hyperspectral imaging datasets discussed herein are foundational for the advancement of deep learning models. These datasets provide a rich

Table 1
The state-of-the-art surveys in HSI data analysis for agricultural with highlighting the research gaps in this field.

Reference	Applications	DL	Description	Research Gaps
(Aasen et al., 2018), 2018	✓	×	Addressing sensor technology, measurement protocols, geometric processing, and radiometric calibration in relation to the UAV spectral remote sensing techniques. Reflects on the ongoing revolution in remote sensing, identifying current trends, opportunities, and limitations.	- Establishing and communicating standard procedures to ensure data quality. - Understanding the full sensing process to mitigate undesired effects during data acquisition and processing. - Need for knowledge dissemination on sensor technology, data acquisition protocols, and data processing.
(Zhang et al., 2020a), 2020	✓	×	Discussing the advancements in hyperspectral technologies for plant disease detection, focusing on the applications and analysis of HSI. Highlighting the primary challenges in the field and suggests areas for further research.	- Challenges in identifying different pathogens and distinguishing between biotic and abiotic stresses. - Early warning of plant diseases. - Satellite-based hyperspectral technology development. - Need for more precise agricultural informatization development tools.
(Terentev et al., 2022), 2022	✓	×	Delving into the latest developments in hyperspectral remote sensing as a tool for the early identification of plant diseases, this analysis covers four specific crop species. It sheds light on the significant shortcomings within this scientific domain and outlines potential solutions, all the while highlighting the substantial promise HRS holds for the early recognition of plant ailments.	- Strong reliance on plant physiology and biochemistry for waveband interpretation. - Need for a general experimental method to achieve consistent results in plant disease detection using HRS.
Ours, 2023	✓	✓	Dive into the latest proposed Deep learning approaches that have been devoted to HSI data analysis in the agricultural field, with giving more attention to examining and analyzing the effectiveness of these methods using prominent land cover datasets.	- Identification of the most effective DL techniques for agriculture. - Examination of the latest applications of DL techniques in agriculture.

array of spectral information. This wealth of data is instrumental in training models to discern fine details and complex patterns, enhancing their analytical precision and decision-making capabilities. Furthermore, these datasets serve as a rigorous testing ground, enabling the evaluation and refinement of model performance under varied and challenging scenarios. In essence, these datasets fuel the development of sophisticated deep learning algorithms and ensure their applicability and reliability in real-world agricultural contexts.

4. Deep learning approaches for HSI analysis

In the recent decade, Deep learning methods have dominated most of the research tasks in computer vision, including the hyperspectral imaging field; this due to their built-in mechanical capabilities for information representation that reflects steps like feature extraction (Sun et al., 2021a), selection (Zhao et al., 2019), and reduction of spatial-spectral and contextual features (Xu et al., 2019).

In this section, the most recent DL techniques for HSI analysis, including CNN, SAE, RNN, GAN, DBN, TL, SSL, FSL and AL, are presented and elaborated in detail below. Fig. 1 displays the annual distribution of documents retrieved from Scopus using the keywords (Hyperspectral imaging, deep learning, and agriculture) for the period from 2017 to 2023. The graph indicates an initially modest number of publications in 2017, reflecting the emerging nature of applying DL techniques to HSI within the agricultural sector. Thereafter, a discernible surge in research activity has been evident since 2018, highlighting the burgeoning interest in leveraging DL to analyze HSI datasets in agriculture. This upswing can be attributed to several factors: the burgeoning cross-disciplinary collaboration among computer scientists, agronomists, and remote sensing specialists has spurred pioneering studies at the intersection of these fields. The proliferation of advanced DL methodologies, combined with the enhanced accessibility to HSI data, and the introduction of powerful computational resources and accessible open-source deep learning libraries such as TensorFlow (Abadi et al., 2015), PyTorch (Paszke et al., 2017), and Keras [?], have significantly lowered the barriers to entry for researchers wishing to deploy complex models with minimal time or resource investment. The peak observed in 2023 can be ascribed to the reduction in costs and physical footprint of hyperspectral sensors, along with strides in their operational efficiency. These improvements have enabled the incorporation of hyperspectral sensors into drones and portable field devices, allowing for immediate in-situ analysis, which has broadened their practicality for both research and applied uses in agriculture.

sensing experts, fostering innovative research at the confluence of these disciplines. Furthermore, the development of the recent deep learning approaches alongside the availability of hyperspectral data, powerful machine and open source deep learning frameworks for example, TensorFlow (Abadi et al., 2015), PyTorch (Paszke et al., 2017), and Keras (Paszke et al., 2017), made it easier for researchers to implement complex algorithms without significant development time or resources. The zenith in 2022 corresponds to a reduction in the cost and size of hyperspectral sensors, coupled with enhancements in efficiency. This advancement has facilitated their integration into drones and on-field devices for real-time monitoring, thereby expanding their applicability in both research and practical agricultural contexts.

4.1. Convolutional neural network (CNN)

CNNs represent a groundbreaking advancement in the field of machine learning, especially in the processing and interpretation of visual data (Bougourzi et al., 2020). Deriving inspiration from the human visual cortex's functioning, CNNs emulate human visual information processing. Their efficacy is particularly notable in image classification (Blaschke et al., 2014), object detection (Ren et al., 2015), and segmentation tasks (Bougourzi et al., 2023a, 2023b), and they play a pivotal role in HSI. The distinct architecture of CNNs is tailored for the

Table 2
Comparison of different HSI agricultural datasets with Source, Spatial Dimensions(SD), Spectral Bands(SB), Wavelength (WL), Spatial Resolution (SR).

Dataset	Year	Source	SD(pixels)	SB	WL(nm)	Samples	Classes	SR(m)
Indian Pines (Scenes, 2020)	1992	NASA AVIRIS	145 x 145	220	400–2500	10249	16	20
Salinas (Scenes, 2020)	1998	NASA AVIRIS	512 x 217	224	360–2500	54129	16	3.7
Pavia University (Scenes, 2020)	2001	ROSIS-03 sensor	610 x 610	115	430–860	42776	9	1.3
Botswana (Scenes, 2020)	2004	NASA EO-1	1496 × 256	242	400–2500	3248	14	30
Chikusei (Yokoya and Iwasaki, 2016)	2014	Headwall Hyperspec-VNIR-C imaging sensor	2517x2335	128	363–1018	77592	19	2.5
WHU-Hi-HanChuan (Zhong et al., 2018)	2016	Headwall Nano-Hyperspec imaging sensor	1217 × 303	274	400–1000	257530	16	0.109
WHU-Hi-HongHu (Zhong et al., 2018)	2017	Headwall Nano-Hyperspec imaging sensor	940 × 475	270	400–1000	386693	22	0.043
WHU-Hi-LongKou (Zhong et al., 2018)	2018	Headwall Nano-Hyperspec imaging sensor	550 × 400	270	400–1000	204542	9	0.463

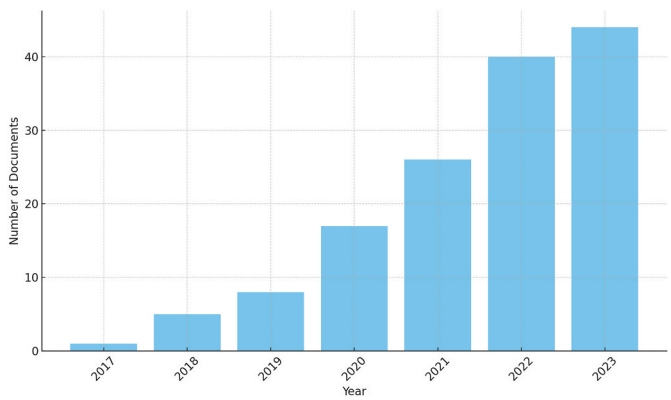

Fig. 1. Number of published articles by year of HSI in the agriculture sector, (source: Scopus).

efficient handling and interpretation of image data. Fig. 2 shows a graphical representation of a CNN architecture.

The architecture encompasses various layers, each serving a specific purpose. The Convolution Layer is fundamental, primarily functioning in the extraction of features from input images (LeCun et al., 2015). This is achieved through the application of a collection of filters or kernels. These traverse across the image, performing element-wise multiplication with the overlapping segments. This operation, known as convolution, produces a feature map highlighting specific characteristics of the input, such as edges, textures, or colours (Krizhevsky et al., 2012). Particularly in hyperspectral imaging, convolution layers are crucial for extracting spatial and spectral features across different wavelengths. Subsequent to the convolution layers is the Pooling Layer, whose purpose is to reduce the spatial dimensions—namely, the width and height—of the input volume for the layers that follow. A common technique used here is max pooling, where the largest element from a designated region of the feature map is chosen. This approach not only reduces computational demands but also helps prevent overfitting by providing a more abstract representation of features. It additionally imparts invariance to the features against changes in scale and orientation.

CNN have gained popularity in LULC classification for their superior proficiency in processing hyperspectral images, extracting spectral-spatial discriminative features as evident in several studies (Lin et al., 2016; Wambugu et al., 2021). It has been repeatedly shown that CNNs exceed the performance of conventional machine learning techniques like Random Forest (RF), Support Vector Machine (SVM), and k-Nearest Neighbors in numerous datasets (Carranza-García et al., 2019). In (Yin et al., 2023), a novel approach utilizing a superpixel model with multi-scale CNNs has been proposed, aiming at improved high-level feature extraction. Employing a multi-scale CNN for every superpixel, this methodology has adeptly addressed the issue of misclassification caused by varying scales in detailed LULC classes, proving to be superior to the method that uses a single-scale CNN for each superpixel, as evidenced by the results of (Chen et al., 2019a). A further significant development in this domain is the unveiling of a Deep Hybrid Dilated Residual Network (DHDRN) (Cao and Guo, 2020). This innovation was assessed using three publicly accessible datasets, benchmarking it against leading methods. The results of these experiments have demonstrated that DHDRN achieves higher classification accuracy and efficiency, eclipsing former approaches. Specifically, the DHDRN technique has showcased exemplary classification accuracies, achieving 99.26%, 99.44%, and 97.96% on the Pavia University, Indian Pines, and Salinas datasets, respectively. Through detailed ablation studies, the authors have illuminated the integral role of specific components, with the hybrid architecture and dilated convolutional layers identified as pivotal to the method's success. The Deep Residual Network (ResNet) (He et al., 2016) has marked a critical development in CNN evolution. Its recent applications in HSI, such as classification (Zhong et al., 2017), denoising (Yuan et al., 2018), spatial resolution enhancement (Wang et al., 2017), and unsupervised feature learning (Mou et al., 2017a), underscore its versatility and effectiveness.

Despite the fact that CNNs have proved their efficiency for hyperspectral imaging analysis, they still have many limitations. First and foremost, they often require huge amount of labelled data for training, which can be expensive to obtain for hyperspectral datasets due to the need for domain-specific expertise. Moreover, hyperspectral data inherently suffer from the curse of dimensionality, consisting of hundreds of spectral bands, leading to computational and memory

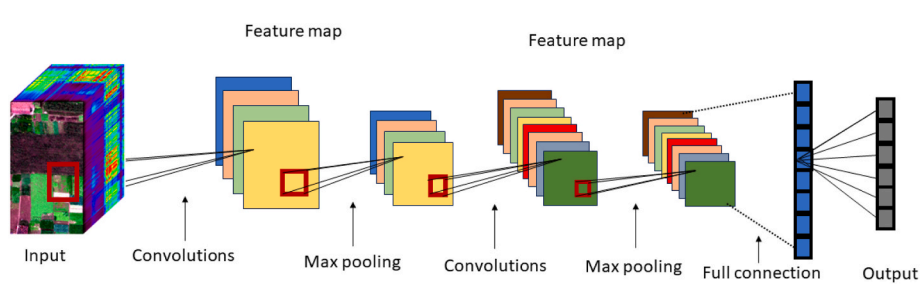

Fig. 2. Graphical representation of CNN architecture. The process starts with the input image, followed by multiple layers of convolutions and max pooling, which serve to extract and condense features. Subsequent layers progressively refine these features, which are then fed into fully connected layers that integrate the information. The final output layer provides the classification result based on the analyzed features. This architecture enables the CNN to learn from visual data and make predictions or classifications effectively.

challenges when processing such high-dimensional data. Additionally, CNNs, with their intricate architectures, demand substantial parameter tuning and computational resources, potentially diminishing their suitability for real-time applications or deployment in resource-limited environments.

4.2. Stacked auto-encoder (SAE)

SAEs have emerged as a valuable and versatile tool in HSI, offering a unique approach to feature extraction and representation learning from high-dimensional hyperspectral data. In essence, SAEs are a type of artificial neural network that learns to encode input data into a lower-dimensional latent space and then decode it back to its original form, effectively compressing and reconstructing hyperspectral information. Their application in HSI has yielded significant contributions to the state of the art. SAEs in HSI can reduce data dimensionality while preserving essential spectral and spatial information. By automatically learning relevant features from hyperspectral cubes, SAEs facilitate data compression, making it more manageable for subsequent analysis tasks, such as classification (Lin et al., 2013), and target detection (Xie et al., 2019; Han et al., 2014). SAEs have demonstrated superior performance in unsupervised feature learning (Tao et al., 2015), enabling the discovery of intricate spectral signatures and spatial patterns that might be challenging to discern. Fig. 3 depicts an example of three stacked autoencoders SAE. The last classification layer of the supervised model to fine-tune is usually achieved by logistic regression or softmax layer.

Furthermore, SAEs have been instrumental in addressing the issue of limited labelled data in HSI. In scenarios where obtaining labelled samples for training DL models is arduous and costly, SAEs can pre-train layers of a deep network in an unsupervised manner, then fine-tune them with a small amount of labelled data as in Fig. 3. This semi-supervised approach has been invaluable in achieving competitive classification accuracy even with limited labelled samples. In 2016, (Xing et al., 2016), the SAE model was deployed to derive functional high-level features essential for remote sensing image categorization. Reconstruction of the input data was accomplished through the use of auto-encoders (AE), which were trained independently and subsequently integrated into each layer of the SAE. The process of extracting abstract features involved a layer-by-layer reconstruction of the input data. During the unsupervised pretraining period, the features learned from one AE informed the progressive training of the subsequent AE, thus streamlining the reduction of reconstruction errors for each AE. After this pretraining phase, the foundational parameters, encompassing weights and biases for all AEs, were established as the preliminary benchmarks for SEA. Refinement of each layer's parameters ensued through error backpropagation, activated by the introduction of labelled data as the supervisory cue, while updates to the structure's parameters leveraged the stochastic gradient descent algorithm. (Vincent et al., 2008, 2010) proposed denoising auto-encoders (DAE) and stacked denoising auto-encoders (SDAE) as other enhancement strategies (Deng et al., 2023). An innovative and solid framework for HSI classification, leveraging a Compact and Discriminative Stacked Autoencoder (CDSAE), is delineated in (Zhou et al., 2019a). This framework is structured around two phases. The first phase involved the extraction of low-dimensional discriminative features through the application of local Fisher discriminant regularization to the hidden layers of the SAE. Following this, the integration of an adept classifier enhanced the training sequence. Tested across three diverse HSI datasets, the introduced technique demonstrated unparalleled superiority in the accurate and consistent classification of HSI. Other approaches investigate the use of two-stage SAEs to find better spectral-only representation with 1D convolutions on input datacube to reduce dimensionality (Bai et al., 2023). Then, a second SAE finds optimal spectral-spatial representations with the reduced spectral data and uses a hybrid auto-encoder that implements 2D and 3D convolution operations to fine-tune a classification task.

Pixel decomposition for subpixel information extraction is another essential research direction in this context where AE and SAE have been proposed in (Yang et al., 2023a) for the *Unmixing* task. The hyperspectral unmixing task decomposes the measured pixel spectrum of the input datacube into a collection of constituent spectral signatures (or *endmember*) and a set of corresponding fractional abundances. An optimal endmember is characterized as a distinct, theoretical, and unadulterated spectral representation of a surface material. The retrieval of stable and preferred endmembers is crucial for attaining precise classification of hyperspectral data and effective spectral separation. The abundance map (Williams et al., 2017) identifies the proportion of each endmember present in the spectra of each pixel. Usually, endmembers are extracted using geometric assumptions, while in (Yang et al., 2023a), a spatial-spectral attention-based bilateral AE network for unmixing is proposed. The attention mechanisms are based on the Swin transformer (Liu et al., 2021a) and were effective in acquiring global spatial attention to extract abundance maps and endmembers of water, soil and trees of the Samson dataset.

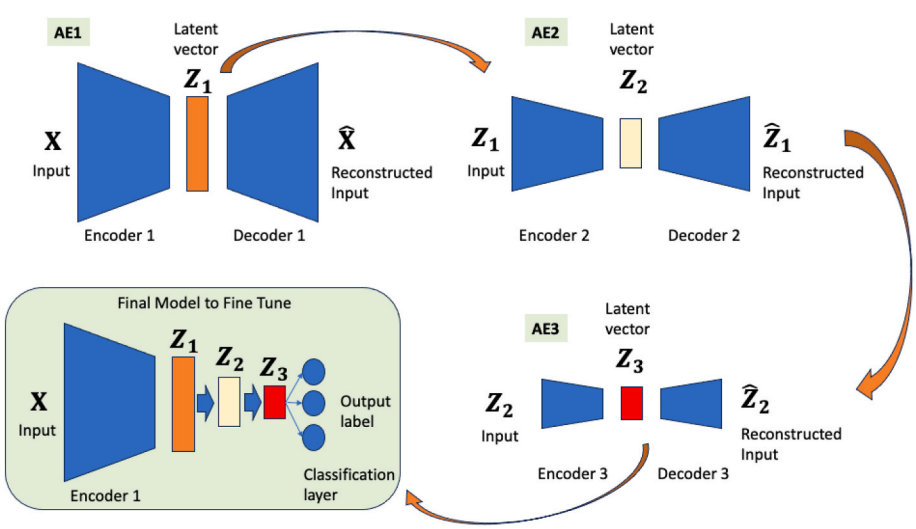

Fig. 3. Example of SAE with three stacks AE1, AE2 and AE3 independently trained in an unsupervised way. The first AE1 is trained to reconstruct the input signal X. The second AE2 is trained to reconstruct the latent variable Z_1 of AE1, and AE3 is trained to reconstruct the latent variable Z_2 and generate a third data representation Z_3. Then, a model that takes the three latent variables to solve a classification task in a supervised way is fine-tuned.

However, the application of SAEs in HSI also presents its own set of challenges. One primary concern is the selection of appropriate architecture and hyperparameters, as these can significantly impact the quality of learned features and the model's overall performance. Like other DL techniques, SAEs require substantial computational resources for training and inference, posing challenges for real-time applications or resource-constrained environments. Interpretability remains an issue, as the complex, hierarchical representations learned by SAEs can be challenging to interpret in hyperspectral data analysis, particularly in scientific and regulatory contexts where model transparency is crucial.

4.3. Deep belief network (DBN)

DBNs have emerged as a promising alternative in HSI, offering a robust framework for unsupervised feature learning and complex spectral and spatial information representation. In essence, DBNs (Hinton, 2009) are a type of artificial neural network with multiple hidden layers, which allow them to capture hierarchical and abstract features from hyperspectral data, making them a valuable tool for extracting meaningful information from these high-dimensional datasets. Within this structure, every hidden layer is constituted as a Restricted Boltzmann Machine (RBM) (Ackley et al., 1985), and the features identified by each layer of RBM are treated as the precursor input for the subsequent layer. Conclusively, the parameters across the network undergo meticulous fine-tuning through a supervised method, integrating a softmax classifier in the terminal layer. DBNs have the ability to automatically discover and model intricate spectral signatures and spatial patterns within hyperspectral data. By iteratively learning representations at different levels of abstraction, DBNs excel in feature extraction, which is crucial for various tasks such as classification (Li et al., 2014), target detection (Zhang et al., 2020b), and anomaly detection (Zhang et al., 2022a). They have effectively handled hyperspectral data's complex, nonlinear relationships, improving classification accuracy and data interpretation. Fig. 4 represents a schematic illustration of the DBN.

Furthermore, DBNs have shown promising results in Semi-Supervised Learning (SSL) scenarios where labelled data is limited or expensive to obtain. They can pre-train layers of the network in an unsupervised manner, capturing essential features from the data distribution, and then fine-tune the model with a small amount of labelled data, thus mitigating the label scarcity issue common in HSI applications. This semi-supervised approach has the potential to significantly improve the efficiency and effectiveness of hyperspectral data analysis.

In (Gavade and Rajpurohit, 2020), the Firefly Harmony Search Deep Belief Network (FHS-DBN) model was employed for LULC classification across four standard datasets. Training of the DBN involved a combined methodology that merges the Firefly Algorithm (FA) with the Harmony Search (HS) algorithm, resulting in the formulation of the FHS algorithm. This methodology yielded encouraging outcomes in LULC classification, underscoring the efficacy of amalgamating various optimization techniques within Deep Learning (DL) models to enhance their performance.

An interesting approach to the application of DBN in HSI is proposed in (Li et al., 2019), where the data reduction is addressed through Principal Component Analysis (PCA) on source hyperspectral data. After this, the authors analyze the performance of a classifier by varying the input features: they compare the use of only Spatial features and both Spectral and Spatial ones. The reduced data are linearized in a 1-dimensional feature vector and provided to a DBN composed of three hidden layers. The approach has been tested on both Indian Pines and Pavia University datasets, and the combination of Spectral and Spatial information outperforms the use of only spatial data.

The applicative context of the approach proposed in (Sun et al., 2021b) differs from those examined up to now. Here, the goal is to detect lead pollution in lettuce leaves, and this approach starts from a laboratory setup where a hyperspectral sensor examines the leaves. The authors compare the results obtained using DBN with those obtained through a conventional approach, which combines Competitive Adaptive Reweighted Sampling (CARS) for input data linearization and Support Vector Machine (SVM) for classification. The DBN is composed of three RBMs: one input layer, two hidden layers, and one output layer. The input layer size was selected according to the input image size, while the number of output layers was set to four according to the number of output pollution levels. DBN outperforms CARS + SVM and provides promising results in evaluating lead pollution, opening interesting applications of DBNs in food quality control.

However, the application of DBNs in HSI also faces several challenges. One key concern is the computational complexity of training deep networks, especially when dealing with large hyperspectral

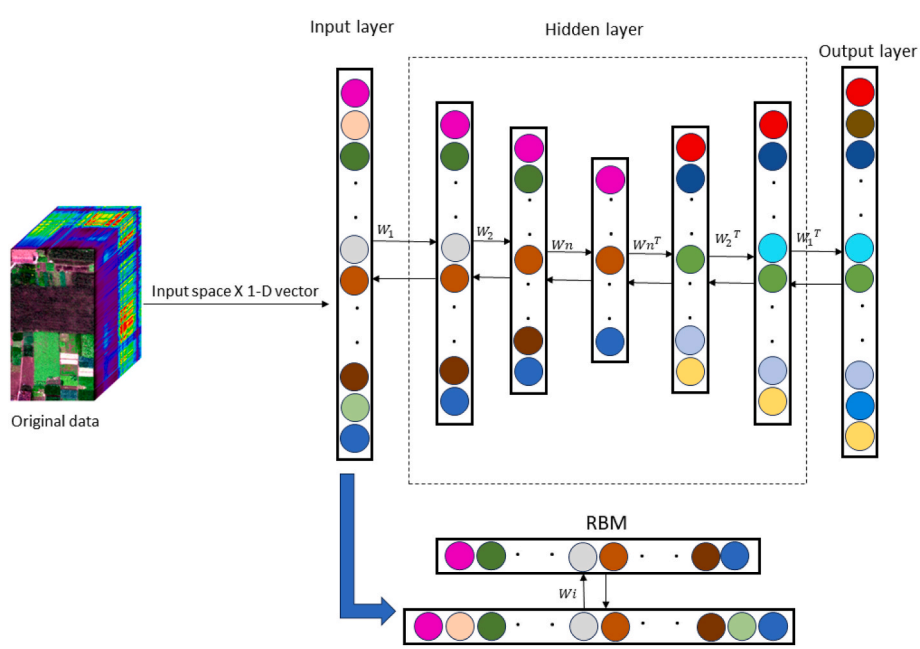

Fig. 4. Diagrammatic illustration of DBN, it is composed of multiple Restricted Boltzmann Machines RBMs with connections between the layers, where the output of a layer is used as the input of the next layer. The W value resresents the weight matrices between the layers of the DBN.

datasets. The need for substantial computational resources can limit their practicality in real-time applications or resource-constrained environments. Additionally, like other DL techniques, DBNs may suffer from issues related to interpretability, making it challenging to understand the reasoning behind their decisions in critical scientific or regulatory contexts. Moreover, the success of DBNs often depends on the availability of appropriate hyperparameters and architectural choices, which can be a non-trivial task in practice. A primary challenge with DBNs is their reliance on random weight initialization. In (Li et al., 2022a), a strategic layered initialization technique is introduced, leveraging the local geometric arrangement for network initialization. Deep features extracted from the multi-DBN framework are then used to develop a discriminative manifold layer. This involves a graph-based strategy for penalty allocation, designed to elucidate the manifold structures within the HSI data's deep features. The evaluations on the Indian Pines, Botswana, and Salinas datasets indicate impressive potential. Additionally, the times required for computation are substantially lower than those recorded for similar benchmarks.

4.4. Recurrent Neural Networks (RNN)

RNNs have emerged as a dynamic and adaptable tool in HSI, offering a unique approach to analyzing the spectral dimension of hyperspectral data cubes. In essence, RNNs are a type of artificial neural network with loops that allow them to process sequential data, making them well-suited for hyperspectral time-series analysis. RNNs in HSI can capture temporal dependencies within hyperspectral data. By considering the sequential nature of data acquisition in HSI, RNNs excel at modelling and exploiting the dynamic characteristics of the scene being observed. This makes them particularly valuable for tasks such as change detection, where detecting subtle alterations in a scene over time is crucial, and for tracking the movement of objects or phenomena across multiple spectral bands. Fig. 5 represents a graphical representation of RNN. Moreover, RNNs have demonstrated effectiveness in hyperspectral data denoising and restoration. Their recurrent architecture allows them to learn and leverage the spectral correlations present in the data to reduce noise and enhance the quality of hyperspectral images. This has critical applications in remote sensing, where atmospheric interference and sensor noise can degrade the quality of acquired hyperspectral data.

The application of RNNs in HSI classification has gained prominence due to their proficiency in processing hyperspectral pixels as sequential data (Mou et al., 2017b). However, the efficacy of standard RNNs is often hampered by critical issues like the explosion or disappearance of gradients. This setback has been significantly addressed with the development of advanced models such as Long Short-Term Memory Networks (LSTMs) (Jiang et al., 2018) and Gated Recurrent Units (GRUs) (Chung et al., 2014), which facilitate the unravelling of the spectral data across a graph, thereby obviating the requirement for recurrent operations. Different LSTM models, such as LSTM-F (Kuchaiev and Ginsburg, 2017), LSTM-S (unidirectional) (Gers et al., 2000), and the base LSTM (b-LSTM)-S (bidirectional) (Huang et al., 2015) aims to solve the challenges of sliding-window segmentation and operates across multiple modes. Utilizing the Bidirectional-Convolutional Long and Short-Term Memory Network (Bi-CLSTM) facilitates the autonomous learning of spectral-spatial features from hyperspectral data, enhancing classification results by about 1.5% in comparison to 3D-CNNs (Liu et al., 2017). Furthermore, a novel HSI classification method, known as Spectral-Spatial LSTM (SS-LSTM), is proposed in (Zhou et al., 2019b), incorporating CNNs and Principal Component Analysis (PCA) for the extraction of spectral and spatial attributes from HSI. These attributes are subsequently processed through separate LSTM layers to analyze spectral sequences and their interconnections. The SS-LSTM model is rigorously trained on an extensive HSI dataset and tested on three benchmark datasets.

However, the application of RNNs in HSI is not without challenges. One primary concern is the potential computational complexity, especially when dealing with high-resolution hyperspectral data cubes, which can be memory-intensive and time-consuming to process. Additionally, selecting an appropriate RNN architecture and optimizing hyperparameters can be challenging, as the effectiveness of RNNs is sensitive to these choices. Interpretability of RNNs in hyperspectral data analysis also remains an issue, as understanding the learned temporal patterns and their relationship to specific spectral features can be complex.

4.5. Generative Adversarial Networks (GAN)

GANs have emerged as a transformative and innovative approach in HSI, introducing a dynamic interplay between two neural networks, the generator and the discriminator, to create, manipulate, and enhance hyperspectral data. In essence, GANs can generate synthetic hyperspectral images that closely resemble real-world data, opening new

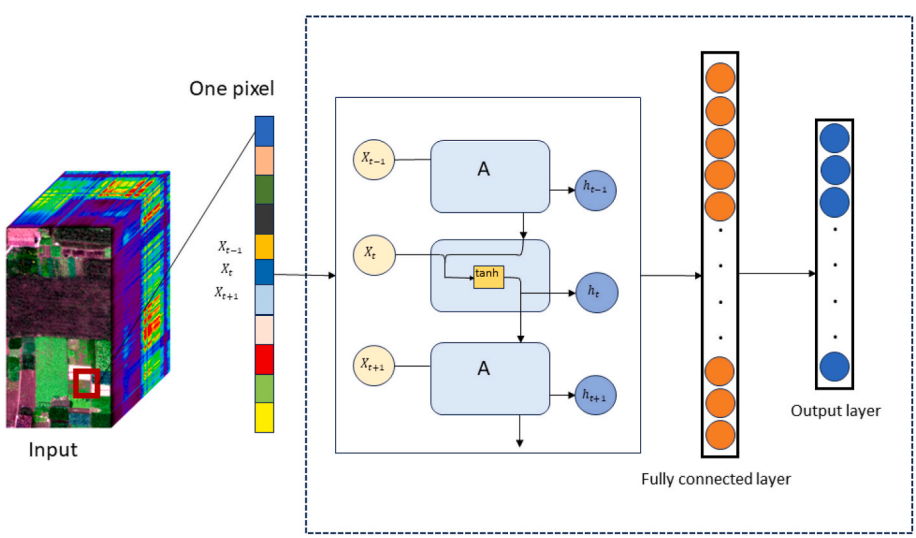

Fig. 5. Graphical representation of RNN architecture. The input cube is decomposed into sequences where each pixel spectrum is treated as a timestep. RNN cells, denoted by "A", process these sequences, utilizing the tanh activation function for non-linearity and maintaining memory across timesteps through recurrent connections. The output from the RNN cells is fed into a fully connected layer, culminating in an output layer designed to deliver the final analysis or classification result based on the hyperspectral data's spatial and spectral features.

alternatives in data augmentation (Zhang et al., 2022b), denoising (Zhang et al., 2022c), and super-resolution (Shi et al., 2022). Hyperspectral data acquisition is often costly and resource-intensive, resulting in small labelled datasets. GANs can help mitigate this challenge by generating synthetic data samples that expand the training set, enhancing the performance of subsequent machine learning models, including classification and anomaly detection algorithms (Deecke et al., 2019). Fig. 6 represents a schematic illustration of the GAN.

Additionally, GANs have shown promising results in hyperspectral data denoising. The discriminator network can effectively learn to distinguish between noisy and clean hyperspectral images, while the generator network strives to produce images that deceive the discriminator. This adversarial process can lead to the generation of denoised hyperspectral images that retain critical spectral information while reducing noise and artefacts, thereby improving data quality for downstream analysis. Moreover, GANs have been used for hyperspectral super-resolution, enhancing the spatial resolution of hyperspectral data. By learning the relationship between low-resolution and high-resolution hyperspectral images, GANs can generate super-resolution images that reveal finer spatial details, benefiting various applications such as target detection, vegetation monitoring, and mineral identification (Abdollahi et al., 2020), (Maimaitijiang et al., 2021).

Several improved GANs, including Deep Convolutional GAN (Chen et al., 2019b), 1-D and 3-D GAN (Zhu et al., 2018), Capsule GAN (Xue, 2020), Cascade Conditional GAN (Liu et al., 2020a), MDGAN (Gao et al., 2019), and 3DBF-GAN (He et al., 2017), have been utilized for HSI. GANs have shown promising results with a small number of labelled samples by fully exploiting sufficient unlabelled samples (Zhan et al., 2017). In (Zhao et al., 2023), a novel approach involving GANs for analyzing HSI data is detailed. Here, the primary HSI dataset undergoes a random division into categories for training, validating, and testing purposes. Training images are subsequently processed through the GAN network, which is designed to understand the inherent distribution and generate artificial samples. The process culminates with the application of PCA to the augmented training set, aiming to reduce its dimensionality. The output data are transmitted to a Multi-Scale Residual Attention network (MSRA) for training. The network parameters are constantly updated using the validation set to obtain the best training model. The experiments obtained by the authors demonstrate that augmenting the training samples using GANs can effectively improve HSI classification accuracy.

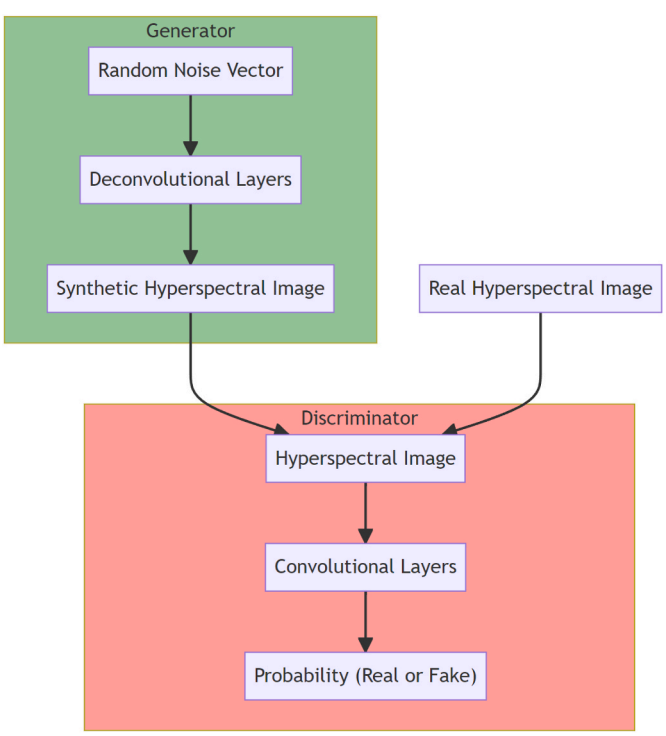

Fig. 6. Schematic representation of GAN architecture. The Generator starts with a random noise vector and transforms it through deconvolutional layers to produce a synthetic hyperspectral image that mimics the statistical properties of real hyperspectral data. Concurrently, the Discriminator receives both real hyperspectral images and the synthetic images produced by the Generator. It uses convolutional layers to process these inputs and outputs a probability indicating whether the given hyperspectral image is real or fake. The generator creates images so convincingly that the discriminator cannot distinguish them from actual hyperspectral images, while the discriminator aims to classify real and synthetic images accurately.

In (Zhang et al., 2023), the strategic use of GANs to circumvent the limitations imposed by small sample sizes and uneven sample distribution is demonstrated through the introduction of the Feature-Preserving Generative Adversarial Network Data Augmentation (FPGANDA) strategy. This technique is notable for its conservation of the primary spectral bands in HSI data via a selective banding process. Moreover, it crafts synthetic spectral bands that are integrated with the original spectral bands using a sophisticated merging approach, thereby enriching the dataset. The evaluation and comparative studies were conducted using the datasets from Indian Pines, the University of Pavia, and Salinas Valley.

Various experts maintain that the assessment capability of an individual discriminator limits the quality of generated samples. In (Gao et al., 2019), a novel strategy is proposed that utilizes a scoring mechanism from a multi-discriminator (MDGAN) collaboration paired with extensive semi-supervised classification. Compared with the standard GAN architecture, this updated evaluation criterion proves to be more precise, with the generated samples displaying significantly sharper characteristics. The technique underwent testing across three datasets, offering a detailed insight into the impact of the number of discriminators on the effectiveness of classification results.

Addressing data reduction is the core of the approach proposed in (Martinez et al., 2023), which is called Low-Dimensional GAN (LD-GAN), in which a pre-trained autoencoder network is used to make a reduced representation of the dataset that will be used for the training of GAN. The obtained low-dimensional samples are then mapped to the spectral image dimension with the pre-trained decoder network. In order to preserve the variability of the generated samples, a statistical regularization approach is also introduced. Similarly, in (Byju et al., 2023), the authors focus on the spatial-spectral redundancies by introducing an algorithm to perform spatio-spectral compression of HSI.

In (Feng et al., 2023), a multi-complementary GANs with contrastive learning (CMC-GAN) has been introduced where it takes the use of two groups of GANs: the first one is demanded to a coarse-grained GAN and adopts the structure in encoder-decoder form for hidden fine-scale and coarse-scale generation, while the second one is a fine-grained GAN, responsible for fine-scale generation.

In the method proposed by (Zhu et al., 2018), the fusion of a 1D-GAN for spectral classification and a 3D-GAN for spectral-spatial classification forms the foundation of the approach. Adversarial samples produced by both GANs are combined with genuine training samples to optimize the discriminative CNN, leading to improved classification outcomes. The effectiveness of this approach is evidenced by the results from experiments conducted on three established datasets.

Despite the fact that GANs have shown their efficiency in dealing with different challenges concentrating on HSI data, they still face specific challenges. First is the potential for mode collapse, where the generator produces a limited variety of samples, leading to a lack of diversity in the generated data. Hyperparameter tuning and the choice of network architectures are crucial to mitigate this issue. Additionally, GANs require substantial computational resources and training time. The generation of the data will be only in the training phase, so it does not have any relationship with the inference time (real-time applications) and resource-constrained environments. Lastly, ensuring the

generated data's quality and fidelity to real-world hyperspectral data remains a challenge, as GANs can introduce subtle artefacts or biases that may impact the reliability of subsequent analyses.

4.6. Discussion

DL techniques have shown significant promise in HSI. These methods excel at extracting intricate features from hyperspectral data, offering advanced capabilities. However, they also present challenges such as computational complexity, the need for extensive labelled data, and interpretability concerns. Despite these hurdles, with appropriate optimizations, DL holds immense potential to revolutionize the HSI domain. Table 3 provides an analytical comparison of advancements in DL classification approaches between 2016 and 2023, utilizing prominent HSI datasets.

Over the years, a variety of methodologies have been explored. The presented Overall Accuracy (OA) and Average Accuracy (AA), show a general trend of improvement over the years. For instance, in 2016, the OA for IP was 80.66%, while by 2023, it reached as high as 99.16%. This progression underscores the rapid advancements in DL techniques for classification tasks over the span of these years. The distribution of literary studies analyzed for each selected DL technique is shown in Fig. 7 as a pie-chart with percentage values for each category.

The chart is dominated by the CNN method, which occupies a staggering 88.1% of the total. This dominance underscores the effectiveness and popularity of CNNs in processing and analyzing the high-dimensional data typical of hyperspectral imaging. Such networks are adept at capturing spatial and spectral features. On the other hand, RNNs and GANs, represented by 5.3% and 6.1%, respectively, indicate their emerging roles, possibly in sequence-based hyperspectral data analysis and data augmentation or synthetic data generation. DBN holds the smallest portion at 2.6%, suggesting its niche application or potential underutilization in this domain.

5. Learning techniques in HSI

In this section, we explore several techniques that have been pivotal in advancing HSI classification, each of which leverages different strategies to enhance learning from limited samples. These techniques, which include Active Learning (AL), Transfer Learning (TL), Semi-Supervised Learning (SSL), and Few-Shot Learning (FSL), are designed to use available data to build robust classification models intelligently. Through the continuous evolution of these techniques, achieving remarkable feats in HSI classification has been possible even with limited labelled data.

5.1. Active Learning (AL)

Active learning (AL) is a paradigm where the algorithm dynamically selects the data points from which it learns. This selection process is guided by a criterion that aims to choose the most informative samples. The idea is to enable the model to achieve higher accuracy with fewer training samples compared to passive learning, where the model is trained on a randomly selected dataset without any input on the selection process (Brame, 2016). This approach is beneficial in scenarios where labelled training samples are limited, helping to improve the quality of training samples and reduce the costs associated with obtaining many labelled training samples. In (Falahatnejad and Karami, 2023), they introduced a classification technique for HSIs based on augmented AL. It involves a two-step process, including feature extraction and classification, using an augmented AL technique. Additionally, in (Yadav et al., 2022), authors proposed a novel AL approach called Pixel-Certainty Active Learning (PCAL) to enhance the classification performance of land cover classes. The methodology involves preprocessing steps such as Distributed Intensity Filtering (DIF) and Histogram Equalization (HE) (Distante and Distante, 2020) to improve

Table 3
Comparison analysis of DL Classification Techniques. IP: Indian Pines, PU: Pavia University, SA: Salinas.

Year	Article	Methodology	Results
2016	Zabalza et al. (2016)	Bands segmentation using SSAE, local SAEs to process original features	IP: OA-80.66% PU: OA-97.5%
2018	Singh and Kasana (2018)	LPP, Deep features using SAE	IP: OA-84.4%. SA: OA-87.2%.
2018	Paoletti et al. (2018)	3DCNN	IP: OA 98.37%, AA 99.27% PU: OA 98.06%, AA 98.61%.
2019	Zhou et al. (2019b)	Spectral-spatial LSTM, PCA	IP: OA 95.00%, k 94.29% PU: OA 98.48%, 97.56% SA: OA 97.89%, k 97.65%.
2020	Vaddi and Manoharan (2020)	Data normalization and CNN-based classification	IP: OA 99.02%, AA 99.17% PU: OA 99.94%, AA 99.92%.
2020	Cao and Guo (2020)	3D-2D spectral spatial hybrid dilated residual networks	IP: OA 99.46%, k 99.38% PU: OA 99.81%, k 99.74%.
2021	Singh and Kasana (2021)	SAE, PCA and LPP-based spectral-spatial features	IP: OA 99.21%, AA 99.19% PU: OA 99.89%, AA 99.90%
2022	Zhan et al. (2022)	Combination of LSTM, residual network and spectral-spatial attention network	IP: OA 97.69%, AA 97.19% PU: OA 95.87%, AA 95.37% SA: OA 98.34%, AA 98.84%.
2022	Pande and Banerjee (2022)	Self-looping CNN	PU: OA 77.76%, AA 71.36% SA: OA 99.32%, AA 99.68%.
2022	Yadav et al. (2022)	Combination of differential texture pattern extraction and PCAL	IP: OA 97.6% PU: OA 98.48%.
2022	Li et al. (2022b)	Termed manifold-based multi-DBN (MMDBN)	IP: OA 81.50% SA: OA 91.79% BO: OA 94.06%.
2023	Alkhatib et al. (2023)	Multi-scale 3D-CNN and three-branch feature fusion	PU: OA 92.66%, AA 90.65% SA: OA 96.68%, AA 97.69%.
2023	Zhou et al. (2023)	Shallow-to-Deep Feature Enhancement SDFE with CNN	IP: OA 99.16%, AA 99.07% PU: OA 99.80%, AA 99.70%, SA: OA 99.97%, AA 99.80%.
2024	Chen et al. (2024)	Center Pixel in Transformer for Hyperspectral Image Classification	IP: OA 98.87%, AA 98.07% PU: OA 98.10%, AA 97.30%,

(continued on next page)

Table 3 (*continued*)

Year	Article	Methodology	Results
			SA: OA 99.17%, AA 99.10%.

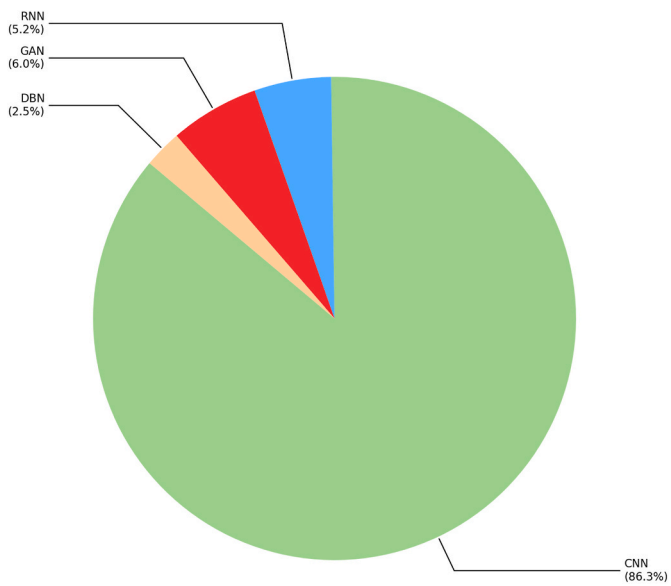

Fig. 7. Pie-chart of related articles on DL approaches used for HSI classification (source: Scopus).

image quality, followed by feature extraction using an extended differential pattern (EDP) to obtain textural information. The PCAL technique leverages this information to classify HSI patterns, demonstrating robustness to pixel noise variations.

Despite its potential, AL faces several challenges, including the computational complexity and resource intensity of processing and analyzing hyperspectral data, which can be prohibitive for real-time applications (Aljundi et al., 2022). Active learning models also rely on the initial selection of samples, and a poor initial selection can significantly impact the learning efficiency and accuracy of the model. These algorithms may have difficulty dealing with the inherent noise and variability in hyperspectral images, affecting their ability to discriminate between different materials or objects accurately.

5.2. Transfer learning (TL)

Through transfer learning (TL) strategies, pivotal information from source data is seamlessly integrated into target data, substantially lowering the reliance on training samples. This enhancement is facilitated by the conveyance of optimally adjusted parameters from source to target data, which in turn, initializes network parameters based on those from established deep networks. This practice yields improved classification results over those obtained through random initialization of network parameters.

"In the architecture of deep networks, the superficial layers are tasked with acquiring elementary features. The network is pre-trained using an appropriate dataset that provides a sufficient quantity of labelled instances, facilitating the transplantation of parameters from the lower and middle layers to a subsequent network that shares a similar structure. Transfer learning (TL) fundamentally encompasses two strategies: 1) the fine-tuning approach, and 2) the method of freezing and then training.

Fine-tuning is a method where a pre-trained network on source data is adapted for target data by engaging in comprehensive layer training. It demands a designated volume of labelled samples for model formulation but demonstrates improved classification performance relative to simpler classifiers. The research documented in (Xie et al., 2021) unveils a spectral-spatial HSI classification approach that incorporates SP pooling CNN and leverages TL by fine-tuning. The findings confirm that this technique is effective in classifying HSI, notwithstanding a minimal number of training labels.

Freezing and training is a technique that trains the classifier on deep features sourced from the transfer network, freezing every layer except the ultimate one for the training phase. Additionally, this strategy might involve immobilizing the initial layers and adjusting the later ones. It employs datasets that are closely related to the target data for pre-training the DL network, pinpointing mutual features across source and target datasets. In (Li et al., 2022c), the classification approach includes pre-training a CNN on a dataset with labels and choosing a limited number of training samples from a distinct dataset. This method freezes the parameters of the early convolutional layers while retraining the more advanced convolutional and fully connected layers. By freezing particular layers, the model keeps the features it learned during pre-training, utilizing existing insights and reducing the need for computational resources during training. Meanwhile, retraining specific layers helps the model to identify and learn features particular to the new dataset, thereby boosting its efficacy in the new classification challenge.

In (Roy et al., 2023), the introduction of the Multimodal Fusion Transformer (MFT) network is detailed, incorporating a multi-head cross-patch attention (mCrossPA) mechanism specifically designed for HSI LULC analysis, drawing on the complementary insights from additional multimodal datasets. The MFT model seeks to improve generalization through the use of tokenization, creating classification and HSI patch tokens for a unique feature representation in a minimized feature space. Extensive comparative analyses with leading-edge models on benchmark datasets revealed that the MFT model outperforms others, primarily due to the innovative application of the mCrossPA mechanism.

Despite its advantages, TL can sometimes lead to negative migration and over-adaptation due to differences in the data distribution between the source and target domains. It also faces challenges when the source and target data have different dimensions, necessitating innovative solutions like iterative reweighting of heterogeneous TL frameworks.

5.3. Semi-Supervised Learning (SSL)

SSL is proposed as a middle ground between supervised and unsupervised classification, aiming to extract useful features from a large number of unlabelled data. This approach is vital in HSI classification research, where hyperspectral labelled samples are hard to obtain. The goal is to design robust and effective SSL frameworks to improve classification accuracy. In (Zheng et al., 2022), an SSL-based preprocessing solution called *NearPseudo* is proposed. In this approach, pseudo-labels are created by an initialization classifier and added to minority classes along with the corresponding unlabelled samples. A feedback mechanism based on a consistency check is introduced to enhance reliability and reduce the misclassification cost of pseudo-labels. The method demonstrated improved classification accuracy compared to other common processing methods and can be applied flexibly to most typical classifiers.

While SSL offers a viable pathway to enhance HSI classification, especially in scenarios with limited labelled training samples, it faces substantial challenges. These include dealing with HSI data's nonlinearity and high correlation, handling high-dimensional datasets, and the continuous requirement for algorithm optimization. Addressing these challenges is crucial to furthering the effectiveness of SSL in HSI classification, and it remains a hot spot in current research endeavours.

5.4. Few-Shot Learning (FSL)

FSL is a strategy that utilizes a small number of labelled samples to train the model to recognize objects. The primary objective is to study the differences between samples rather than learning what the sample is directly. It operates on the principle of making the model "learn to learn.". In (Yang et al., 2023b), the proposed technique Multipath and Multiscale Siamese Network (MSSN). The network leverages spatial-spectral features and has low dependence on sample information, aiming to enhance classification performance with few-shot training samples. In (Xia et al., 2023), a Convolutional Residuals and SAM Siamese Network (CRSSNet) was proposed to enhance feature extraction and mitigate network overfitting and gradient degradation issues in few-shot hyperspectral image classification. The method demonstrates superior classification performance compared to other methods.

FSL, while promising, faces several challenges that can impede its effectiveness. One of the primary issues is the ease with which it can produce invalid samples, affecting the classification results and increasing computational time. This learning approach is also susceptible to negative migration and over-adaptation, exacerbated by the often different data distributions between the source and target domains. Moreover, FSL sometimes requires too frequent human-computer interactions. This process can be further complex when synthetic samples generated are meaningless and cannot be labelled, hindering learning. Another significant challenge is the need for improvements in training within a high-dimensional parameter space, a requirement that can potentially lead to increased space and computational costs. Addressing these challenges is crucial to harnessing the full potential of FSL in HSI classification, focusing on efforts to mitigate issues related to data validity, adaptation, and computational demands.

6. Applications of HSI technology in agriculture

This section illustrates and outlines the most significant contributions of HSI in agriculture applied to private datasets.

6.1. Soil analysis

Crop growth relies on the vitality of the soil environment. Rapid and precise access to data on soil nutrient composition is vital for targeted fertilization practices. Neglect in soil management poses a threat to soil's functionality and its role in agricultural production, especially in rural economies (Song et al., 2018). Climatic variations across regions play a significant role in shaping soil conditions and vulnerability to erosion, necessitating the identification of soil erosion for addressing agricultural dilemmas, such as enhancing crop yield and quality. Through the application of HSI technology, soil analysis is revolutionized, enabling the collection of comprehensive data over a broad electromagnetic spectrum. The analysis of such imagery permits the identification and quantification of soil attributes, including levels of soil moisture (Wu et al., 2012), nutrients (Jia et al., 2017), and mineralogy (Manley, 2014). This crucial information supports the mapping of soil traits, facilitating informed agricultural and environmental management decisions. The search continues for more efficient, non-destructive approaches to detect soil organic matter (SOM), fundamental to the dynamics of the soil-plant ecosystem.

In (Wei et al., 2021), UAV hyperspectral data over Daye City in China is used, a mining suspected contaminated area, to establish a model for retrieving Total Arsenic (TAs). They collected 74 soil samples in situ and measured their Total Arsenic (TAs) contents using Atomic Fluorescence Spectrometry (AFS). This study unveiled a novel method that integrates deep neural networks with competitive adaptive reweighted sampling (DNN-CARS) for the determination of soil total arsenic (TAs) levels and their spatial dispersion. The experiments show that the combination of UAV hyperspectral imagery and DNN-CARS is highly effective in measuring TAs levels and charting their spatial layout.

In (Singh and Kasana, 2022), a novel framework for estimating soil properties was proposed. Emphasizing the significance of soil property estimation for sustainable agriculture, the authors leverage the sequential nature of hyperspectral signatures using hybrid features and RNN variants (LSTM and GRU). Principal Component Analysis (PCA) and Locality Preserving Projections (LPP) were used to reduce the dimensionality of hyperspectral data. The proposed methods demonstrated superior performance in predicting soil properties such as pH of H_2O, Nitrogen, Organic Carbon, and Sand compared to existing models.

HSI offers detailed spectral information for soil analysis, but it faces several challenges. The high dimensionality of HSI data makes processing and storage computationally demanding. Soil's inherent heterogeneity, combined with the effects of moisture, makes the interpretation of spectral signatures difficult. Atmospheric interference, especially when data is acquired from airborne or satellite platforms, requires corrections. This is justified by the variations in illumination and shadows that can skew spectral responses. Proper sensor calibration is crucial, and any drift can introduce inaccuracies. The vast data volume from HSI poses storage and processing challenges, and the high spectral resolution can sometimes introduce noise. Novel algorithms for analysis can be computationally intensive, and the overall cost of high-resolution hyperspectral sensors and tools can be prohibitive for some applications. Table 4 summarizes some of the most important research on HSI for soil analysis applications.

6.2. Crop yield estimation

Yield forecasting stands as a pivotal field within precision agriculture research. Essential activities such as managing crops, matching crop supply with market demand, forecasting yields, mapping yields, and charting crop distribution are critical for maximizing agricultural output (Al-Gaadi et al., 2016). Addressing the challenge of estimating crop production is among the most effectively resolved issues through precision farming techniques. By leveraging HSI technology, it becomes

Table 4
An overview of the key research conducted on soil analysis using HSI.

Objective	Article	Method	Results
Rapid identification of microplastics in farmland soil	Ai et al. (2022)	DT, SVM, CNN	CNN accuracy = 92.6%, DT accuracy = 87.9%, SVM accuracy = 85.6%
Geographical origin identification of Chinese chestnuts	Li et al. (2021)	1D-CNN	1D-CNN accuracy = 97.12%
Quantitative estimation of soil properties	Singh and Kasana (2022)	Hybrid features with LSTM and GRU	Improved R^2 values for multiple soil properties; RMSE reduced by up to 1.76 times; Highest RPIQ of 6.67 for pH(H2O)
Estimation of soil properties	Singh and Kasana (2022)	Hybrid features, LSTM	$R^2 = 0.85$, RMSE = 10.56
Extracting Soil Moisture	Wang et al. (2020)	DBN with FY-3D MERSI-II imagery	RMSE values: LR (0.101), BP neural network (0.083), SM-DBN (0.032); SM-DBN outperformed other models
Estimating spatial distribution of soil TAs	Wei et al. (2021)	Deep Neural Networks with Competitive Adaptive Reweighted Sampling (DNN-CARS)	R^2 values of 0.90 and 0.87; RMSE values of 0.33 and 0.52; Effective tool for evaluating TAs content

possible to gauge crop yields by analyzing the vitality and health of the vegetation, which includes data on chlorophyll levels, hydration, and nutrient status. This wealth of data enables accurate crop yield estimations, anticipates crop stress and potential decreases in yield, and refines crop management methods, all of which contribute to yield improvement. Furthermore, by capturing comprehensive spectral insights from the visible to near-infrared parts of the electromagnetic spectrum, HSI technology plays a crucial role in identifying and mapping specific vegetation types and changes in the terrain over time, enhancing the precision of yield estimations.

In (Pang et al., 2020), a distinction between seed vigour detection and prediction technique was proposed. The study evaluates the potential of HSI technology and convolutional neural networks (CNNs) to identify and predict maize seed vitality. The hyperspectral information of four vigor level seeds was collected 10 h before germination. Various models, including a support vector machine, an extreme learning machine, and a one-dimensional convolutional neural network (1DCNN), were used to model the spectral dataset. The 1DCNN outperformed other models on the original spectral data, achieving a recognition accuracy of 90.11%. The research also highlighted the importance of the first 3 h of data for prediction, which showed higher recognition accuracy than the test set. An image-based 2DCNN model achieved a remarkable 99.96% accurate recognition. By differentiating the spectra and image information, the CNN models can achieve accurate detection and prediction, offering a promising framework for advancing research on seed germination. The paper emphasizes the potential of combining HSI with DL techniques to revolutionize the process of seed vitality detection and prediction, making it more efficient and accurate. Additionally, in (Wen et al., 2023), high-resolution HSI data was captured at various growth stages of corn, which was subsequently denoised using wavelet analysis. This imagery was then utilized for training and validating a CNN model, incorporating both spectral and colour image data. The results showcased that the integrated CNN model, which combined both types of data, achieved a classification accuracy of 75.50%. This accuracy was notably higher than models that relied solely on either spectral or colour image information.

Estimating crop yield using HSI presents several challenges. Crop canopies are inherently heterogeneous, with varying leaf orientations, densities, and health conditions, leading to diverse spectral responses. Atmospheric interference, particularly from airborne or satellite platforms, requires meticulous corrections. Shadows, varying illumination, and the presence of other elements in a pixel, like soil or water, can confound the crop's spectral signature. Calibration drifts in sensors can introduce inaccuracies, and the immense data volume from HSI demands robust storage and processing solutions. While HSI can capture physiological changes, distinguishing those relevant to yield from other stressors is a challenge. Additionally, the high cost of hyperspectral sensors and the computational intensity of advanced analysis algorithms can be limiting factors for widespread use in crop yield estimation. Table 5 summarizes the most important research on HSI for crop yield estimation applications.

6.3. Agricultural crop classification

One significant area of study for several agricultural applications is the identification and classification of the crop using hyperspectral images. HSI technology can assist in agricultural crop classification by capturing detailed spectral information of crops in the visible and near-infrared regions of the electromagnetic spectrum. This information can identify unique spectral signatures for different crop types and growth stages, which can be used for classification purposes. Machine learning algorithms are typically used to analyze hyperspectral data and classify crops based on spectral reflectance characteristics. This information can create maps of crop types and growth stages, providing valuable agricultural management and decision-making insights. Additionally, HSI can be used to detect crop stress (Römer et al., 2012), diagnose potential yield-limiting factors (Vigneau et al., 2011), and monitor crop health, which are all important for crop management and optimization. HSI can also be used to monitor changes in the landscape over time, allowing for the detection of crop growth and yield changes and providing a valuable tool for precision agriculture. Table 6 summarizes some of the most significant research on HSI for agricultural crop classification applications.

Table 5
An overview of the key research on the application of HSI for crop yield prediction.

Objective	Article	Method	Results
Prediction of corn yield	Yang et al. (2021)	2DCNN	Classification accuracy = 75.50%, Kappa coefficient K = 0.69
Accurate predictions of wheat yields	Cheng et al. (2022)	LSTM, RF, GBDT, SVR	LSTM model performed best with RMSE of 0.201 t/ha
HSI for corn yield estimation	Yang et al. (2021)	CNN, Wavelet denoising	Integrated CNN model achieved 75.50% accuracy
Prediction of Corn Seeds	Pang et al. (2020)	CNN(1D and 2D) + HSI	(1DCNN) Classification accuracy = 90.11%, (2DCNN) Classification accuracy = 99.96%
Classify of different crop types	Farmonov et al. (2023)	WA-CNN	OA = 97.89% and classification accuracy from 97% to 99%

6.4. Contaminants and nutrient estimation

One of the significant applications of HSI is in the detection of contaminants on various surfaces, especially in food products (Ravikanth et al., 2017; Barbin et al., 2013). Contaminants can range from foreign materials, such as plastic or metal fragments, to biological contaminants like mould or bacteria (Rather et al., 2017). Since different materials have unique spectral signatures, HSI can identify these contaminants by analyzing the spectral information of each pixel. For instance, a piece of plastic on a leafy vegetable might reflect light differently than the surrounding leaves. By analyzing the spectral data, HSI can pinpoint the exact location and nature of the contaminant. In addition to detecting contaminants, HSI is also used for estimating nutrient content in food and agricultural products (Zhang et al., 2013). Different nutrients, such

Table 6
An overview of the most significant studies conducted on agricultural crop classification using HSI.

Objective	Article	Method	Results
Corn seed variety classification	Zhang et al. (2021)	DCNN, KNN	DCNN classification accuracy = 100% Testing accuracy rate = 94.4%, 57 and validation accuracy rate = 93.3%
Variety identification of coated maize kernels	Zhang et al. (2020c)	LR, SVM, CNN, RNN and LSTM	Classification accuracy over 90%
Prediction of intact oranges consistency parameters	Riccioli et al. (2021)	ANN	(SSC)RMSECV = 0.87% (TA) RMSECV = 0.23 RMSECV = 2.78 for MI RMSECV = 1.11 for brima
The inherent uniformity of apple fruit slices is accentuated	Lan et al. (2021)	Near-infrared HSI	Rcv2 = 0.83, TSC Rcv2 = 0.81, RPD = 2.20 and RPD = 2.39
Predict the viability of pepper seeds	Hong et al. (2023)	HSI PLS-SVM, Xray Ensemble-SVM	PLS-SVM OA = 88.99%, Ensemble-SVM OA = 92.51%
Improving Green Pepper Segmentation	Liu et al. (2023)	CVNN based on 1D Fast Fourier Transform	Classification accuracy = 94.89%, F1 Score = 89.55%

as vitamins, minerals, and sugars, interact with light in specific ways, leading to distinct spectral signatures (Liu et al., 2021b). By analyzing the hyperspectral data, scientists and researchers can estimate the concentration of these nutrients in the sample. For example, the ripeness and sugar content of a fruit can be determined by examining its spectral response. This application is particularly valuable in agriculture, where farmers can use HSI to monitor the nutrient content of their crops and make informed decisions about harvesting and post-harvest processing.

However, when it comes to estimating contaminants and nutrients, several challenges arise. The complexity and variability of natural samples can lead to overlapping spectral signatures, making it difficult to distinguish between different substances. Additionally, the presence of external factors like moisture, lighting conditions, and surface roughness can influence the spectral response, potentially leading to inaccurate estimations. The high dimensionality of hyperspectral data also poses computational challenges, requiring robust algorithms and models for effective data processing and interpretation. Furthermore, calibration between laboratory and field measurements is crucial, as discrepancies can lead to significant errors in estimation. Thus, while HSI holds promise for contaminants and nutrient estimation, addressing these challenges is essential for its effective application. Table 7 summarizes some of the most significant research on HSI for estimating contaminants and nutrients.

6.5. Plant disease monitoring and invasive plant species

The appearance of several pests and illnesses in the crops poses serious challenges to farmers. Some frequent causes of illness infections include nematodes, bacteria, viruses, and fungi. Due to unawareness of crop diseases and the need for expert support and advice, farmers have historically avoided diagnosing or suspecting most infections. Disease infections should be detected early on to prevent crop damage. HSI technology can assist in plant disease monitoring and invasive plant species detection by analyzing the reflectance spectra of the plants in multiple narrow, contiguous wavelength bands. This technology can detect subtle changes in the chemical and physical properties of plants that are not visible to the human eye (Moghadam et al., 2017), such as changes in chlorophyll content or leaf water content. This information can then be used to identify plant stress and disease symptoms (Mahlein et al., 2012), such as discolouration or wilting, or to distinguish between invasive plant species and native species based on differences in their spectral signatures. This allows for early detection and monitoring of plant diseases and invasive species, which can lead to improved management strategies and outcomes.

Invasive plants often disrupt local ecosystems, outcompeting native species and affecting biodiversity. Early detection and accurate identification are key to controlling these invasive populations. HSI offers a powerful method for distinguishing invasive species based on their unique spectral signatures. While two plant species may look similar to the human eye, their spectral profiles captured across hundreds of wavelengths may differ significantly, providing the means for definitive identification.

Machine learning and DL algorithms have started to play a pivotal role in plant disease monitoring and invasive species identification. Complex algorithms are being trained to recognize the spectral signatures associated with particular diseases or invasive plants, automating the identification process. This is particularly useful in large-scale monitoring, where manual inspection would be time-consuming and impractical. Additionally, machine learning and DL models can adapt to new information, becoming more accurate and potentially recognizing new diseases or invasive species as they appear. Table 8 summarizes some of the most significant research on HSI for Plant disease monitoring and invasive plant species applications.

6.6. Plastic pollution

Even if not strictly correlated with agriculture, plastic pollution has become one of the most emergent issues threatening aquatic and terrestrial ecosystems, so its implications in agriculture are intrinsic. The detection of plastic in the wild is a challenging area. As evident, the size of the monitored plastics is related to the resolution of the sensor, i.e. the flight altitude of the UAV. Microplastic can be detected only in the laboratory, with a distance between the target and the sensor in the range of centimetres (Zhu et al., 2021) or online with the latest digital

Table 8
Overview of the most significant research conducted in the field of agricultural applications of HSI for the monitoring of plant diseases and invasive plant species.

Objective	Article	Method	Results
Detection of target spot and bacterial spot diseases in tomato	Abdulridha et al. (2020)	MLP Neural Network, Stepwise Discriminant Analysis (STDA)	Classification accuracy = 99% for both target spot (TS)
Classification of the asymptomatic biotrophic phase of PLB disease	Qi et al. (2023)	2DCNN and 3DCNN with attention networks	Classification accuracy = 79.0% F1 score = 0.83
3D DL for plant disease recognition	Nagasubramanian et al. (2019)	DCNN	Classification accuracy = 95.73%, F1 score = 0.87
Identification of red-berried wine grape infected with grapevine leaf-roll disease	Gao et al. (2020)	Monte-Carlo, SVM, GLD	Classification accuracy = 89.93%
Aflatoxin B1 detection	Zhu et al. (2023)	Dual-branch CNN	Classification accuracy 91.30%
Assessment of weed competitiveness in maize farmland ecosystems	Lou et al. (2022)	3D-CNN	RMSE = 0.106 and 0.152 using 13 feature bands
Disease detection of basal stem rot	Yong et al. (2023)	Mask RCNN and VGG16	Classification accuracy 91.93%

Table 7
An overview of the most significant research on the use of HSI for estimating contaminants and nutrients.

Objective	Article	Method	Results
Estimation of wheat nitrogen content	Al Makdessi et al. (2019)	PLS-R	Average error = 0.83%DM
Assessment of chlorophyll content	Gao et al. (2021)	ROI-ALR- RF -PLSR with ROI-OMR	Rv2 = 0.52 and RMSE = 3.61
Estimation of the starch content in an individual kernel	Liu et al. (2020b)	ANN, PLSR, adaptive reweighted sampling	Rp2 = 0.96 RMSEP = 0.98
Estimation of nitrogen content in leaves of olive trees	Rubio-Delgado et al. (2021)	PLSR, SVM, LNC, SM	R2 = 0.72, R2cv = 0.71, R2 = 0.64, Rp2 = 0.63
Various minerals types classification	Agrawal and Govil (2023)	Mineral–CNN–LSTM based on 1D-CNN and LSTM and mineral-ResNet based on 1D-CNN, LSTM	mineral-ResNet OA = 92.16% and kappa value of 0.89, mineral–CNN–LSTM OA = 91.71% and kappa value of 0.88

holographic microscope technology (Bianco et al., 2020), (Cacace et al., 2023). On the other hand, macroplastics can be detected at a higher distance; basically, many approaches use satellite images (Themistocleous et al., 2020), (Topouzelis et al., 2020). These approaches suffer from the main limitations of this kind of sensor: low spatial resolution, fixed time samples, rigid protocols for data access, and no customizable acquisition campaigns. On the other hand, they can count on high-quality data regarding spectral resolution. The target of such approaches is large plastic waste detection. It is performed using some specific spectral indices such as Reversed Normalized Difference Vegetation Index (RNDVI), Normalized Difference Water Index 2 (NDWI2), Plastic Index (PI), and Floating Debris Index (FDI), as reported in (Page et al., 2020) and (Biermann et al., 2020).

The possibility to mount a hyperspectral sensor on a UAV opens a large field of applications oriented to small plastics detection (Cortesi et al., 2022). In this scenario, the limitations of satellite data are overcome, and the increase in spatial resolution makes a comfortable contribution to developing on-demand models able to perform path planning, data acquisition, and data processing in a compact time lapse. These advantages are indisputable and contribute to the growing use of these methodologies. In (Gonçalves and Andriolo, 2022) and (Balsi et al., 2021), authors propose an approach to detect marine litter by using limited spectral bandwidth samples. Random forest classifiers can perform the processing of acquired data (Cortesi et al., 2022) or, recently, by DL-based approaches (Maharjan et al., 2022). Using information from a sensor working in Short-Wave InfraRed (SWIR) bandwidth is proposed in (Cocking et al., 2022). This spectrum area is mainly used to distinguish between different plastics (i.e. PET, PVC, and so on). At the same time, the detection can also be performed in different bandwidths; specifically, the range $600 - 900$ nm is suitable for this. In recent years, the use of plastic in agriculture has become massive, mainly due to an increase in plastic-covered greenhouse farming areas and plastic-mulched farmlands; this makes it necessary to develop tools and approaches for the automatic detection and monitoring of such areas for sustainable development of horticulture, including high-quality agricultural production, and reduced pollution (Veettil et al., 2023), (Levin et al., 2007), (Roldán et al., 2015).

6.7. Moisture estimation

Moisture estimation using HSI uses the unique spectral signatures associated with water content (Amatya et al., 2012). Water molecules absorb electromagnetic energy at specific wavelengths. By capturing the reflected or emitted energy across a broad spectrum, hyperspectral sensors can detect the dips or energy peaks corresponding to these water-absorbing wavelengths (Zovko et al., 2017). The DL approaches can then analyze this spectral information to estimate the moisture content of the material being observed. However, several factors can influence the accuracy of moisture estimation. These factors include the quality of the hyperspectral sensor, the presence of other substances that may have overlapping spectral characteristics, and the angle and conditions of light exposure. Furthermore, calibration is essential to link the spectral data to actual moisture levels, often requiring ground-truth data for verification.

Machine learning and DL techniques are increasingly used to enhance the estimation process. These algorithms can be trained on labelled data sets that link spectral information to known moisture levels, improving the model's predictive accuracy. Machine learning models can also account for complex interactions between various spectral bands that may be difficult to capture using traditional analytical approaches.

6.8. Anomaly detection

Anomaly detection in HSI is a sophisticated process that leverages hyperspectral sensors' rich spectral and spatial information to identify unusual or unexpected objects or materials within a scene. The core of this process lies in the ability of hyperspectral sensors to capture image data across a wide range of electromagnetic spectrum, enabling the differentiation of objects based on their spectral signatures. This wealth of spectral information is pivotal in various applications, including vegetation analysis (Cavaliere and Senatore, 2021), geological survey (Yang et al., 2015), and surveillance (Li et al., 2020b). In (Zhuang et al., 2020), the challenge of denoising in HSI, which is crucial for effective anomaly detection, was addressed. The authors proposed RhyDe, a robust HSI denoiser that implements explicit low-rank representation and promotes self-similarity while preserving rare pixels. This approach significantly enhanced the denoising and detection effectiveness, paving the way for better anomaly detection in noisy hyperspectral data. Usually, the anomaly detection framework is based on the deep neural network architecture of autoencoders, which is able to build a compact representation of the input in the latent space that is successively reconstructed with a decoder to get the desired output. It is an unsupervised method, and input-output volumes are usually datacubes temporally adjacent. So far, given an input datacube I_1 at time step t_1, the network is forced to predict the next acquired datacube I_2 at time step t_2. This is usually done with acquired data without annotation. Once the autoencoder is trained to predict the next HSI image, anomalies emerge at the testing phase (change detection) when the expected reconstructed datacube is dissimilar to the actual acquired HSI image. In (Hu et al., 2021), a similar architecture is presented with a siamese autoencoder to model two temporally adjacent HSI. Authors utilized autoencoder networks to model spectral variation and highlight anomaly changes between multi-temporal hyperspectral images. This nonlinear solution proved efficient and superior to traditional methods, especially in complex imaging conditions, thus demonstrating the evolving nature of anomaly detection methodologies in hyperspectral imaging.

7. Discussion

While HSI technology presents significant advancements over traditional nondestructive testing methods, it is imperative to recognize the coexistence and relevance of both approaches in the spectrum of analytical methods. The inaccuracy associated with manual processes (Goller et al., 2021), instrument usage (Ioannou, 2023), and reagent preparation in classical detection methods can be unpredictable, yet these conventional methods still hold value for their reliability and established history, particularly in scenarios where HSI's limitations are pronounced.

The large volume of data generated by HSI is both its strength and its challenge. While the data richness allows for a comprehensive understanding of farmland variations, processing and interpreting this data necessitate advanced computational techniques. Traditional methods, though reliable, often fall short in handling the complexity and volume of HSI data. This is where DL approaches, with their power to handle vast datasets, become indispensable. The ability of DL to autonomously identify patterns and correlations within the data not only streamlines the decision-making process but also ensures that decisions are grounded in robust data-driven insights.

DL methods in HSI vary in their approach and purpose. While CNNs focus on extracting spatial patterns through sliding convolutional kernels, 3D convolutional networks incorporate the spectral dimension, allowing for richer representations that capture spatial and spectral features. Autoencoders, however, are designed to compress and reconstruct data, making them useful for dimensionality and noise reduction. Transfer learning leverages models pre-trained on large datasets to address the limited labelled HSI data challenge. In contrast, GANs are utilized for data augmentation and domain adaptation, promoting model generalization across different hyperspectral conditions.

This review encompasses a comprehensive aggregation of scholarly articles, which were meticulously analyzed and categorized based on

varying emphases. The categorization and summarization of these studies across different sections aim to outline a coherent structure showcasing the evolution of HSI alongside the integration of various methodologies. This progression is notably observed in agricultural applications of HSI enhanced by Deep Learning DL, extending further into diverse fields. The growing engagement of researchers in this area has significantly propelled the advancement of this domain.

7.1. Limitations and future scope

Despite the significant challenges associated with processing high-band data, researchers have made significant progress in this field over the last decade, improving existing techniques and developing new ones. With advancements in technology and the introduction of machine learning, HSI classification has become more accurate than traditional and contemporary state-of-the-art methodologies. As a result, DL has emerged as the most prominent tool for HSI classification in the past decade, leading to a greater exploration of remote sensing and space imagery features.

HSI exploration has opened up new alternatives for research with numerous real-world applications. While HSI spectral bands offer certain advantages over multispectral and RGB imaging, including the fact that multispectral captures images in a relatively small number of spectral bands, which can restrict its ability to distinguish between materials or detect specific chemical signatures. HSI also have limitations and disadvantages. Therefore, it is crucial to address the challenges, some of which are listed below:

- **Data Complexity and High Cost**: The intricacies in processing high-band data, coupled with the substantial financial implications, present a significant barrier. An in-depth discussion on the cost-benefit analysis and potential strategies for cost reduction can provide insights into making HSI more accessible.
- **Challenges in Interpretation Due to Low Spatial Resolution**: Despite the rich spectral information, HSI data often suffers from low spatial resolution, leading to irregularities and interpretational difficulties. Future research could focus on techniques to enhance spatial resolution or develop novel interpretation methodologies that mitigate this issue.
- **Limited Training Samples in Public Datasets**: The scarcity of extensive training samples in publicly available HSI datasets hampers the development of robust classification algorithms. Exploring strategies for synthetic data generation or cross-domain data utilization could be potential solutions.
- **Difficulty in Target Detection**: The unpredictable nature of target and background spectra in HSI complicates the development of efficient target detection algorithms. Future studies could delve into advanced pattern recognition and machine learning techniques to address this challenge.

The techniques discussed below offer researchers new and improved approaches to HSI analysis:

- **Meta-learning in HSI Classification**: The potential of meta-learning, which involves creating algorithms that can generalize across datasets, is relatively unexplored in HSI classification. This area holds promise for developing more adaptable and efficient classification systems.
- **Utilization of Newer Datasets**: Employing recently developed datasets like RIT-18, olive, and grape berry datasets can provide a testing ground for existing and improved techniques. Their use could lead to more reliable and versatile classification methods.
- **Automatic Parameter Selection and Optimization**: Implementing evolutionary techniques or genetic algorithms for automatic parameter tuning in deep learning and other methods could significantly enhance the efficiency and accuracy of HSI classification.
- **Redundancy Diagnosis in Data Cubes**: Addressing the issue of redundant information in HSI data cubes through automatic protocols can lead to improved classification accuracy. This approach would involve identifying and eliminating duplicate information within the bands.
- **Development of Mobile HSI Setups**: Creating cost-effective, compact, and portable HSI systems, including filters, sensors, and lighting sources, is an emerging research area with practical implications.
- **Advances in Pattern Recognition**: Exploring recent advancements in pattern recognition to more effectively and efficiently learn from high-dimensional data cubes in HSI is a promising research direction. This approach focuses on extracting and analyzing spectral signatures for enhanced classification accuracy.

8. Conclusion

This comprehensive review focuses on the advancements in utilizing HSI data in agriculture over the past two decades, specifically emphasizing the integration of artificial intelligence (AI) methods for exploiting this valuable data for automatic and real-world applications. The paper thoroughly assesses various methodologies, offering insights into their performance, addressing research gaps, and highlighting achievements. Notably, it stands out by extensively covering the development of deep learning in this context. The analysis of each method leads to significant conclusions, enhancing the overall novelty of the work. The review also details applications of HSI technology in agriculture, providing a thorough examination of the current research landscape in HSI analysis. Additionally, it introduces recently developed techniques with potential implications for future research in the field.

The depth and granularity of data provided by HSI offer an unprecedented window into the intricacies of farmland variations. From understanding the subtlest changes in soil composition to predicting potential pest infestations, HSI provides a comprehensive canvas of information. However, the large volume and complexity of this data necessitate advanced processing techniques. This is where AI, and more specifically DL, steps in. With its ability to sift through vast datasets, identify patterns, and generate actionable insights, AI acts as the perfect foil to HSI's data-rich outputs. Together, they form a potent combination capable of addressing some of the most pressing challenges in agriculture.

Beyond the technical aspects, there is also a broader socio-economic dimension to consider. As HSI and AI become more entrenched in agriculture, it is imperative to ensure that the benefits of this technological convergence are equitably distributed. Ensuring accessibility and affordability will be key to preventing the creation of a technological divide within the agricultural community.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported in part by the Ministry of Enterprises and Made in Italy with the grant ENDOR "ENabling technologies for Defence and mOnitoring of the foRests" - PON 2014–2020 FESR - CUP B82C21001750005; and in part by FAIR "Future Artificial Intelligence Research" CUP B53C220036 30006 grant number PE0000013. The authors thank Mr. Arturo Argentieri from CNR-ISASI Italy for his technical contribution to the computing facilities.

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